

Fundamental Properties of the Degree

Homotopy Invariance

$H:[0,1] \times \Omega \rightarrow \mathbb{R}^n$ continuous,

$H(t,x) \neq y$ on $\partial\Omega$ for all t

$\Rightarrow \deg(H(t, \cdot), \Omega, y)$ is constant in t

Existence Property

$\deg(f, \Omega, y) \neq 0$

$\Rightarrow \exists x \in \Omega : f(x) = y$

(nonzero degree implies solution exists)

Domain Additivity

$\Omega_1 \cap \Omega_2 = \emptyset$, both open in Ω

$y \notin f(\overline{\Omega} \setminus (\Omega_1 \cup \Omega_2))$

$\Rightarrow \deg(f, \Omega, y) = \deg(f, \Omega_1, y) + \deg(f, \Omega_2, y)$

Excision Property

$C \subseteq \Omega$ closed, $y \notin f(C)$

$\Rightarrow \deg(f, \Omega, y) = \deg(f, \Omega \setminus C, y)$

(degree unchanged by removing regions not containing solution)